

Course 6031C

Microcontroller & PLC

Diploma in Electrical and Electronics Engineering

POLY PMNA - Polytechnic Study Hub

Course 6031C - Microcontroller & PLC

Diploma in Electrical and Electronics Engineering - Semester 6

Revision 2021 | Program Elective | Credits: 4 | Periods: 60

A downloadable handbook PDF generated from lessons-6031C.html.

Core areas: 8051 architecture, ALP, 8255 PPI, peripheral interfacing, PLC architecture, scan cycle, ladder logic and industrial motor/lighting control.

Overview

Objectives: understand operation, construction and application of microcontroller; acquire programming knowledge of 8051; understand operation, construction and application of PLC; provide basic knowledge in PLC programming.

Prerequisites: Digital Electronics from Analog & Digital Circuits (Semester 3) and basics of programming from Introduction of IT Systems (Semester 1).

CO1: Summarize architecture of 8051 microcontroller. CO2: Develop programs to interface 8051 with internal and external peripherals. CO3: Illustrate features, block diagram and applications of PLC. CO4: Build programs to interface PLC with internal and external peripherals.

Modules 1 and 2

Module 1: 8051 features, comparison with microprocessor, block diagram, CPU, oscillator, ROM, RAM, I/O ports, serial port, timers/counters, interrupt control, register structure, PSW, TMOD, TCON, IE, IP, memory organization, pin functions, interrupts, serial communication and applications.

Module 2: instruction format, addressing modes, instruction set groups, assembly language programs, 8255 PPI block diagram, control word and interfacing of LED, stepper motor and DC motor.

ALP example: MOV A,30H; ADD A,31H; MOV 32H,A; HERE: SJMP HERE. Result is stored at 32H. Use comments and show result location in exams.

Modules 3 and 4

Module 3: PLC introduction, importance, advantages, disadvantages, relay based panel versus PLC based panel, block diagram, major components, I/O devices, operation, scan cycle, applications and selection criteria.

Module 4: PLC programming classification, ladder diagram, rungs, inputs, outputs, power rails, ladder rules, basic I/O instructions, timer/counter, math, logical, data handling and comparison instructions.

Practice ladder programs: logic gates, staircase lighting, go-down lighting, traffic light, DOL starter and star-delta starter.

Rule Bank and Practice

8051 rule: instruction = opcode + operand. Addressing modes decide where the data is located. ALP sequence should be initialize, load data, execute operation, store result and end loop.

PLC rule: scan cycle = input scan, program scan, output scan and diagnostics. Ladder logic uses contacts as conditions and coils as outputs. Use stop, overload and interlock for motor circuits.

Expected questions: draw and explain 8051 architecture; compare microprocessor and microcontroller; explain 8255 PPI; compare relay and PLC panels; explain PLC scan cycle; develop DOL and traffic light ladder programs.

Model Paper and Answers

Model paper: define microcontroller, list SFRs, explain addressing modes, draw 8051 architecture, write ALP for addition, draw 8255 PPI, compare relay and PLC panels, explain PLC scan cycle and develop ladder logic for DOL starter.

Model answer guide: for 8051 draw CPU, ROM, RAM, ports, timers, serial port and interrupts; for ALP show comments and result; for PLC include I/O table, rung diagram, seal-in contact and overload protection.

References: Kenneth Ayala 8051 Microcontroller Architecture Programming and Application; Mazidi et al. The 8051 Microcontroller and Embedded System; Ajit Pal Microcontroller Principle and Application; Raj Kamal Microcontroller Architecture Programming, Interfacing and System Design; NPTEL resources listed in the supplied syllabus.